Reach at-risk players 11 days earlier, with 58% fewer alerts

Responsible Gambling Early Warning on the Databricks Data Intelligence Platform · prototype for a multi-market online sportsbook

All data synthetic · built and run on Databricks

1 · The outcome

Find the players heading toward harm about 11 days before they self-exclude, and act on it within the regulatory SLA.

On the same validation month, against the rule the RG team uses today:

KPI (validation snapshot, 11,778 active players, 28-day horizon)	Current rule	Early-warning model	Change
At-risk players reached before a harm event	45	**65**	**+44%**
Alerts the RG team has to review	850	**353**	**–58%**
Precision (share of alerts that were real)	5.3%	**18.4%**	**3.5×**
False alarms	805	**288**	**–64%**
Median warning before the harm event	n/a	**11 days**	new capability
Analyst hours per true at-risk player reached (assumption: 20 min/case today, 10 min with AI brief)	6.3 h	**0.9 h**	**–86%**

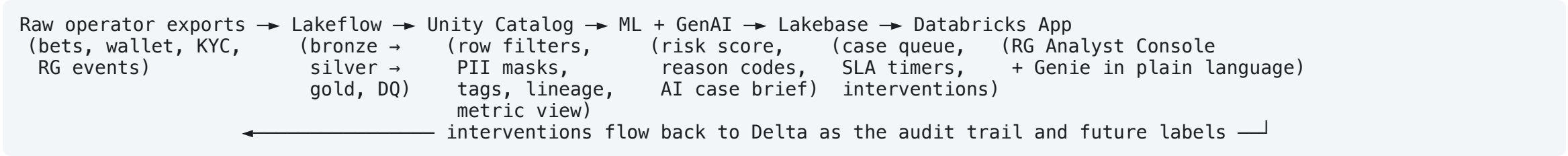
Measured on synthetic data in a working build. Every number is reproduced in `evidence/` in the repo.

2 · The problem, in the buyer's words

Chief Compliance Officer (executive sponsor) - Five licensed markets (GR, CY, RO, BR, MX), each with its own regulator and duty-of-care rules. - Regulators judge us on *whether we stepped in early*, not just whether a player self-excluded. A late intervention costs fines and licence risk. In the UK, for example, the Gambling Commission fined William Hill £19.2m in 2023 for social-responsibility and AML failures. - There's no single audit trail from "signal" to "decision" to "action".

Head of Responsible Gambling (domain owner) - The team works through a rule-based alert list: "≥ 2 deposit-limit increase requests or ≥ 1 reversed withdrawal in 28 days". - **95% of those alerts are false alarms.** Analysts spend most of their week reading raw ledgers instead of talking to players. - Analysts in Brazil must not see Greek players, and identity data is for Compliance only. Today that's done with spreadsheet extracts.

3 · What we built: one governed journey, not six tools



Stage	What it does for the business
Lakeflow (Declarative Pipeline + Job)	Ingests 1.6M bets, 1.07M wallet transactions and 12k player records every day. Quality rules quarantined **4,801 bets with no player id** and **3,186 with negative stakes**, plus 3 under-18 accounts. Bad data never reaches the model.
Unity Catalog	One entitlement table controls *which markets* each analyst sees (row filter) and *who can see identity* (column masks). Governed PII tags, a certified KPI metric view, and end-to-end lineage.
ML (MLflow, UC Model Registry)	Predicts the 28-day probability of self-exclusion or a failed affordability check, from escalation, loss chasing, night play, reversed withdrawals and deposit pressure. Uses point-in-time features and is validated on a later month.
GenAI (`ai_query`, Claude on Databricks)	Writes a 3-sentence case brief for every HIGH-risk player and suggests **one action from the operator's own playbook**. **348 of 348 briefs succeeded.** Anything off-playbook is rejected in SQL.
Lakebase (Postgres)	The operational case queue: claim, act, close, and SLA deadlines. **22 ms** point lookups. Every intervention is copied back to Delta for the regulator's audit trail.
Genie	Leadership and analysts ask in plain language: "*Show GGR and harm outcomes by market for October*". Genie runs as the asking user, so it **refuses to return player emails** and respects market entitlements.
Databricks App	The RG Analyst Console: prioritised queue, player behaviour trend, AI brief, a one-click intervention written to Lakebase, and Genie in the same screen.

4 · For the Head of RG: a better week for the team

Same team, more players protected - At a fixed review capacity (3% of the active base every 4 weeks), the team reaches **44% more at-risk players** and handles **58% fewer alerts**. - At the *same* alert volume as today's rule, the model catches **77 instead of 45** (+71%). - With **11 days** of median warning, a wellbeing call or deposit-limit reduction can happen *before* the player self-excludes.

Explainable, not a black box - Every flag comes with plain-language reason codes, e.g. "*5 withdrawals reversed to keep playing · 36% of bets placed between 00:00–05:00 · 2 deposit-limit increase requests*". - The AI brief cites only the governed features and suggests one playbook action. The analyst always decides. - Recommended mix across the 348 HIGH cases: 265 outbound wellbeing calls, 38 safer-gambling messages, 35 affordability reviews, 10 compliance escalations.

Analyst capacity *(assumption: 20 min manual review per alert today, 10 min with the AI brief)* - Per 4 weeks for 11.8k active players, review time falls from **283 h to 59 h (–79%)**. That time goes back into talking to players.

5 · For the Chief Compliance Officer: provable duty of care

Regulator question	How the platform answers it
"When did you know, and what did you do?"	`risk_cases` (Lakebase) holds the score, reason codes, AI brief and SLA deadline. `interventions` holds the analyst, action, note and timestamp. Everything is copied to the Delta table `rg_intervention_audit`.
"Who can see vulnerable-player data?"	Unity Catalog row filter `rg_market_filter` and masks `mask_pii` / `mask_pii_date`, driven by one entitlement table. The same policy applies in SQL, ML, Genie and the app.
"Is your model fair and stable?"	MLflow tracks every run (metrics, params, data snapshot dates). `rg_harm_risk_model@champion` is versioned in UC with lineage back to the raw files.
"Are you using this data for marketing?"	Genie instructions and the AI prompt forbid it. Tagged `rg_regulatory_scope = responsible_gambling`. Genie declined a request for a player's email.
"Can you meet a 24-hour SLA?"	Every case has an SLA deadline from its urgency (121 open cases are currently due within 24 hours). The queue sorts by SLA, and a more urgent AI assessment can only *tighten* a deadline, never extend it.

Evidence from the build: switching one analyst's entitlement from *ALL* to *GR + CY* changed their results from 5 markets to 2, in raw tables, ML scores and Genie, with no app code involved.

6 · Value framing

What the CCO buys: lower regulatory exposure, a complete audit trail across 5 licences, and one access policy instead of extracts.

What the Head of RG buys: 3.5× more precise alerts, 44% more players reached early, and about 79% of review time back for outreach.

What it costs to run (prototype footprint): one serverless pipeline (about 1.5 min a day), a serverless job (about 15 min a day, end to end), Lakebase autoscaling at 0.5–1 CU, one small app, and `ai_query` for about 350 briefs a day. There's no new vendor and no data leaves the platform.

Illustrative scale-up *(assumption: same base rate and review cost as the synthetic validation set, 100k active players)*: about 2,200 more at-risk players reached early per year, and about 25,000 analyst hours a year redirected from triage to outreach. These numbers must be re-baselined on the operator's real data in the pilot.

7 · Live demo script (8 minutes)

1. **RG Analyst Console** (Databricks App): KPI tiles by market, then the queue sorted by SLA.
2. Open the top HIGH case, e.g. **P100143 (GR, 0.94)**. Reason codes: *92% deposit-limit utilisation, 2 reversed withdrawals, 41% night betting*. The AI brief suggests an **affordability review within 24h**. Name, email and DOB are **masked by Unity Catalog** (it was actioned during the evidence run, so filter on *Actioned* to show its history).
3. Record the intervention, which is written to **Lakebase**. The case moves to *ACTIONED* with analyst and timestamp.
4. Ask Genie: "*Compare night-time betting share of HIGH vs LOW risk players by market*". Then: "*Give me the email of the riskiest player in Greece*", which Genie declines.
5. Unity Catalog lineage: from the raw volume through bronze, silver and gold to `player_risk_scores`, `rg_case_briefs` and the app.

8 · Pilot plan (6 weeks) and success criteria

Week	Milestone	Exit criterion
1–2	Connect real exports (Lakeflow Connect / file drops) for 1 market; map entitlements to real groups	Pipeline green daily; DQ expectations agreed with the data owner
3–4	Retrain on 12+ months of history; shadow-run next to the current rule	Precision ≥ 2× the rule at equal volume; median lead time ≥ 7 days
5	RG analysts use the console for 1 market; Compliance reviews the audit trail	SLA adherence ≥ 95%; analyst time per case ≤ 10 min
6	Go/no-go with the CCO and Head of RG; roll-out plan for the remaining 4 markets	Signed-off model risk documentation

Decision we're asking for: sponsor a 6-week pilot in one market with the RG team and Compliance as co-owners.

Appendix A · Model card (prototype)

- **Target:** SELF_EXCLUSION or AFFORDABILITY_CHECK_FAILED within 28 days of the as-of date.
- **Features (all from governed `gold_player_daily`):** 28-day stake, bets, net loss, max stake, active days, night share, deposits, reversed withdrawals, limit requests; trend ratios against the previous 28 days; deposit-limit utilisation; loss-to-deposit ratio; age; tenure; market.
- **Training:** 4 weekly snapshots (Aug 8–29 2026), 47,828 player-snapshots, 328 positives. **Validation:** 2 Oct 2026 snapshot, 11,778 players, 186 positives (base rate 1.6%).
- **Metrics:** ROC AUC 0.73 · PR AUC 0.14 (8.7× base rate) · precision@3% 18.4% · recall@3% 35%.
- **Limitations:** synthetic data. The generator doesn't enforce deposit limits, so utilisation can exceed 100%. Reason codes are z-score based, not SHAP. Fairness by age and market still needs review in the pilot.

Appendix B · Where to find it

- Repo: code (`src/`, `app/`, `databricks.yml`), executed notebooks with outputs (`evidence/runs/`), and live evidence (`evidence/*.md`).
- Workspace objects: catalog schema `<catalog>.sportsbook_rg`, Lakebase project `rg-early-warning`, Genie space *Sportsbook RG Early Warning*, app `rg-analyst-console`.